Jin Li 李津

Supply chain planning and data, UV-LED water treatment

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Profile

PhD candidate at the University of Tokyo with a year of North America integrated business planning at Lenovo. Cut excess inventory from $12M to $5M, coordinated a China-to-Vietnam motherboard transition through the 2025 tariffs, and built Python automation that raised planning output by 65%. Author of five peer-reviewed papers on drinking-water treatment, two as first author. Works across Python, R, SQL and statistical modelling to turn technical and operational data into decisions.

Experience

Lenovo, Master Planning Specialist (North America IBP)
Sep 2024 – Aug 2025

Global Supply Chain, PC. Hefei and Shanghai, China, hybrid

- Ran North American IBP and end-to-end supply planning for commercial, education and public-sector, and SMB notebooks, balancing demand, supply, inventory, capacity and risk across sales, R&D, procurement, manufacturing and ODM partners. SAP IBP key user.
- Reduced excess inventory from $12M to $5M in six months through life-cycle extensions, part migration, BOM changes, commercial actions and global reallocation, releasing $7M of working capital and cutting warehousing cost by 15%.
- Coordinated the China-to-Vietnam motherboard transition, running both sources in parallel through the new line's ramp and using up China board stock, to cut tariff exposure without a break in supply.
- Pulled North American supply forward and built finished-goods inventory before the 2025 tariff changes, rebalancing other regions and factory capacity.
- Built a Python parts-to-box commit tool with cross-factory material matching, raising planning output efficiency by 65% and reducing origin-related delays.

University of Tokyo, PhD Researcher
Oct 2025 – now

Oguma Lab, Department of Urban Engineering. UV-LED advanced oxidation for drinking water

- Plans and runs end-to-end studies of 280 nm UV-LED/H_2O_2 and UV-LED/chlorine across oxidant dose, pH, temperature and water matrix.
- Developed pCBA probe experiments to quantify hydroxyl-radical exposure (R_{ct}), and a Monte Carlo and Sobol sensitivity workflow to rank matrix effects.
- Tests treatment of coagulated water from a full-scale plant and assesses THM and HAA formation potential; published GSM and 2-MIB removal by UV-LED/HClO (2026).

Dalhousie University, MASc Researcher
Sep 2022 – May 2024

Centre for Water Resources Studies. Partner projects with Halifax Water and AquiSense Technologies

- Managed partner research from requirements and test plans to analysis and reporting; finished the experimental programme ahead of schedule.
- Studied 280 nm UV-LED/chlorine disinfection, ozone and UV-based oxidation, organic matter and by-product formation; compared medium-pressure UV and UV-LED for the utility.
- Assembled and characterised an AquiSense PearlAqua Thinfilm 534A2 flow-through UV-LED reactor and gave design recommendations to the manufacturer.

Magna Mirrors Shanghai, EHS and ESG Engineering Intern
Feb – Apr 2022

- Prepared EHS and ESG reports; supported risk assessment, safety inspections, mirror design updates and communication with international customers.

Weilun Trading, Marketing Intern
Jun 2018 – Sep 2020

- Supported FMCG route-to-market for Weilong, Haidilao and Samyang; SQL demand analysis for peak periods; about 15% lower transport and 10% lower warehousing costs.

Education

PhD, Urban Engineering, The University of Tokyo
2025 – 2028 (expected)

Supervisor: Prof. Kumiko Oguma. University of Tokyo Fellowship, full funding

MASc, Civil Engineering, Dalhousie University
Sep 2022 – May 2024

Water treatment and data analytics. GPA 4.07/4.30, first in programme. Full scholarship. Supervisor: Prof. Graham Gagnon

BEng, Environmental Engineering, Nanjing Forestry University Sep 2018 – Jun 2022
GPA 84/100

Publications

1. **Li, J.**, Tsuruno, H., Kobashi, J., Yoshizawa, K., & Oguma, K. (2026). Effects of temperature and pH on the GSM and 2-MIB degradation by UV-LEDs/HClO advanced oxidation. *Desalination and Water Treatment*, 326, 101746.
2. **Li, J.**, MacIsaac, S., Jiang, X., Anderson, L. E., Rauch, K. D., Prest, T., Swinamer, R., Stoddart, A. K., & Gagnon, G. A. (2025). Combined UV LED and chlorine for synergistic drinking water disinfection and assessment of disinfection by-product formation. *AWWA Water Science*, 7(6), e70046.
3. Bennett, J. L., MacIsaac, S. A., **Li, J.**, et al. (2025). Photolysis at the speed of light: chemical-free degradation of trace organic contaminants by bespoke photolysis using high-intensity UV-C LEDs. *ACS ES&T Water*, 5(11), 6375–6387.
4. Tong, W., Chu, Q., **Li, J.**, et al. (2022). Sequential two-stage pretreatment on lignin properties and holistic utilization of lignocellulose biomass. *Renewable Energy*, 187, 123–134.
5. Pan, Y., Bu, Z., **Li, J.**, et al. (2021). Sulfamethazine removal by peracetic acid activation with sulfide-modified zero-valent iron. *Separation and Purification Technology*, 277, 119402.

One further first-author manuscript under review (2026). Talks: invited oral, IUVA Americas, Halifax, Oct 2026; IUVA Americas 2024; ACWWA 2023; Dalhousie Water Symposium 2023.

Skills

Supply chain	IBP, MRP, SAP IBP, supply-demand balancing, inventory optimisation, capacity planning, network transition, tariff scenarios
Data and modelling	Python, R, SQL, Power BI, Excel, MATLAB, Lingo, Monte Carlo simulation, Sobol sensitivity analysis
Water and lab	UV-LED systems, advanced oxidation, DBP analysis, UV-Vis, fluorescence EEM, HPLC, LC-MS, GC
Languages	Mandarin (native), English (professional, IELTS 7.0), Japanese (JLPT N2, 2026)
Awards	University of Tokyo Fellowship; Dalhousie full scholarship; Challenge Cup university first prize and Jiangsu second prize (2021); two utility model patents